



第四章 作业

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1. 下列数列是否收敛？如果收敛，求出它们的极限：

$$(2)\alpha_n = \frac{i^n}{n} + \frac{1}{n^2}; \quad (3)\alpha_n = (-1)^n + \frac{i}{3^n};$$

解：

(2) $i^n = \{1, i, -i, -1\}$ 有界

$$\lim_{n \rightarrow \infty} \frac{i^n}{n} + \frac{1}{n^2} = 0 \Rightarrow \alpha_n \text{ 收敛}$$

$$(3) \alpha_n = (-1)^n + \frac{i}{3^n}$$

$n \rightarrow \infty$, 实部 $(-1)^n$ 极限不存在,

$\therefore$ 原数列不收敛

3. 判断下列级数的收敛性和绝对收敛性

$$(4) \sum_{n=1}^{\infty} \frac{e^{\frac{i\pi}{n}}}{n};$$

$$(6) \sum_{n=1}^{\infty} \frac{\cos in}{2^n}$$

解: (4) $\sum_{n=1}^{\infty} \left| \frac{e^{\frac{i\pi}{n}}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$, 调和级数发散 $\Rightarrow$ 不绝对收敛

$$\sum_{n=1}^{\infty} \frac{e^{\frac{i\pi}{n}}}{n} = \sum_{n=1}^{\infty} \frac{\cos \frac{\pi}{n}}{n} + i \sum_{n=1}^{\infty} \frac{\sin \frac{\pi}{n}}{n}, \quad \sum_{n=1}^{\infty} \frac{\cos \frac{\pi}{n}}{n} = \cos \pi + \frac{1}{2} \cos \frac{\pi}{2} + \sum_{n=3}^{\infty} \frac{\cos \frac{\pi}{n}}{n}$$

$$\sum_{n=3}^{\infty} \frac{\cos \frac{\pi}{n}}{n} > \sum_{n=3}^{\infty} \frac{\cos \frac{\pi}{3}}{n} = \frac{1}{2} \sum_{n=3}^{\infty} \frac{1}{n} \Rightarrow \sum_{n=3}^{\infty} \frac{\cos \frac{\pi}{n}}{n} \text{ 发散} \Rightarrow \sum_{n=1}^{\infty} \frac{\cos \frac{\pi}{n}}{n} \text{ 发散}$$

$$\frac{\sin \frac{\pi}{n}}{n} < \frac{\frac{\pi}{n}}{n} = \frac{\pi}{n^2}, \text{ 而 } \sum_{n=1}^{\infty} \frac{\pi}{n^2} \text{ 收敛} \Rightarrow \sum_{n=1}^{\infty} \frac{\sin \frac{\pi}{n}}{n} \text{ 收敛}. \therefore \sum_{n=1}^{\infty} \frac{e^{\frac{i\pi}{n}}}{n} \text{ 发散}.$$

$$(6) \sum_{n=1}^{\infty} \frac{\cos in}{2^n}$$

$$\frac{\cos in}{2^n} = \frac{e^n + \frac{1}{e^n}}{2^{n+1}}$$

$$\lim_{n \rightarrow \infty} \frac{\cos in}{2^n} = \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \left(\frac{e}{2}\right)^n + \frac{1}{2^{n+1} \cdot e^n} \rightarrow +\infty \neq 0$$

$\therefore$ 原级数发散

8. 求下列幂级数的收敛半径和收敛域:

$$(4) \sum_{n=1}^{\infty} \frac{1}{n^2 2^n} z^n;$$

$$\begin{aligned} \text{解: } \lambda &= \lim_{n \rightarrow \infty} \left| \frac{n^2 2^n}{(n+1)^2 2^{n+1}} \right| \\ &= \lim_{n \rightarrow \infty} \left| \left(\frac{1}{1 + \frac{1}{n}} \right)^2 \cdot \frac{1}{2} \right| \\ &= \frac{1}{2} \qquad \therefore R = 2 \end{aligned}$$

$|z| = 2$ 时,

$$\sum_{n=1}^{\infty} \left| \frac{1}{n^2 2^n} z^n \right| = \sum_{n=1}^{\infty} \left| \frac{1}{n^2} \right| \text{ 收敛}$$

$\therefore$ 收敛域为 $|z| \leq 2$

9. 求下列幂级数的和函数

$$(2) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n}$$

解:

$$\begin{aligned} & \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n} \\ &= \sum_{n=1}^{\infty} (-1)^{n-1} \int_0^z z^{n-1} dz = \int_0^z \left(\sum_{n=1}^{\infty} (-z)^{n-1} \right) dz \end{aligned}$$

$$|z| \leq 1 \text{ 且 } z \neq -1, \text{ 原式} = \int_0^z \frac{1}{1+z} dz = \ln(1+z)$$

10. 把下列函数展成 z 的幂级数, 并指出它的收敛半径

$$(2) \frac{1}{(1-z)^2}; \quad (4) \sin 2z;$$

解:

$$(2) \frac{1}{(1-z)^2} = \left(\frac{1}{1-z} \right)' = \left[\sum_{n=0}^{\infty} z^n \right]' = \sum_{n=1}^{\infty} n z^{n-1}$$

$$|z| < 1 \quad R = 1$$

$$(4) \sin 2z = \sum_{n=0}^{\infty} (-1)^n \frac{(2z)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} (-1)^n \frac{2^{2n+1}}{(2n+1)!} z^{2n+1}$$

$$|2z| < +\infty \Rightarrow |z| < +\infty \quad R = +\infty$$

11. 把下列函数在指定点 z_0 处展成泰勒级数, 并指出其收敛半径

解: (3) $\frac{1}{z^2}, z_0 = 2$

$$\begin{aligned}\frac{1}{z^2} &= -\left(\frac{1}{z}\right)' = -\left[\frac{1}{2} \cdot \frac{1}{1 + \frac{z-2}{2}}\right]' = -\frac{1}{2} \left[\sum_{n=0}^{\infty} \left(-\frac{z-2}{2}\right)^n \right]' \\ &= -\frac{1}{2} \sum_{n=1}^{\infty} n \left(-\frac{z-2}{2}\right)^{n-1} \cdot \left(-\frac{1}{2}\right) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{2^{n+1}} (z-2)^{n-1}\end{aligned}$$

$$\left| \frac{z-2}{2} \right| < 1 \Rightarrow |z-2| < 2 \Rightarrow R = 2$$

$$(6) \quad e^z, z_0 = 1$$

$$e^z = e^{(z-1)+1} = e^{z-1} \cdot e = e \sum_{n=0}^{\infty} \frac{(z-1)^n}{n!}$$

$$|z-1| < +\infty \Rightarrow R = +\infty$$

12. 把下列函数在指定的环域内展为洛朗级数

解: (4) $\frac{1}{z^2(z-i)}, 0 < |z| < 1, 1 < |z-i| < +\infty;$

在 $0 < |z| < 1$ 内,

$$\text{原式} = \frac{1}{z^2} \cdot \frac{1}{-i(1+iz)} = \frac{i}{z^2} \sum_{n=0}^{\infty} (-iz)^n = \sum_{n=0}^{\infty} (-1)^n i^{n+1} z^{n-2}$$

在 $1 < |z-i| < +\infty$ 内,

$$\text{原式} = \frac{-1}{z-i} \cdot \left(\frac{1}{z} \right)' = -\frac{1}{z-i} \left(\frac{1}{i+(z-i)} \right)' = \frac{-1}{z-i} \left(\frac{1}{z-i} \frac{1}{1+\frac{i}{z-i}} \right)'$$

$$\text{原式} = \frac{-1}{z-i} \cdot \left(\frac{1}{z} \right)' = -\frac{1}{z-i} \left(\frac{1}{i+(z-i)} \right)' = \frac{-1}{z-i} \left(\frac{1}{z-i} \frac{1}{1+\frac{i}{z-i}} \right)'$$

$$= \frac{-1}{z-i} \left[\frac{1}{z-i} \sum_{n=0}^{\infty} (-i)^n (z-i)^{-n} \right]' = \frac{-1}{z-i} \left[\sum_{n=0}^{\infty} (-i)^n (z-i)^{-n-1} \right]'$$

$$= \frac{-1}{z-i} \left[\sum_{n=1}^{\infty} -(n+1)(-i)^n (z-i)^{-n-2} \right] = \sum_{n=1}^{\infty} (-i)^n (n+1)(z-i)^{-n-3}$$

$$\begin{aligned}
 & (6) \sin \frac{1}{1-z}, 0 < |z-1| < +\infty \\
 & = -\sin \frac{1}{z-1} = -\sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{1}{z-1}\right)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} (-1)^{n+1} \frac{(z-1)^{-2n-1}}{(2n+1)!}
 \end{aligned}$$

$$(7) e^{\frac{1}{1-z}}, 1 < |z| < +\infty$$

$$= e^{-\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}}} = e^{-\frac{1}{z} \cdot \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^n} = e^{-\sum_{n=0}^{\infty} z^{-n-1}} = e^{-z^{-1}} \cdot e^{-z^{-2}} \cdot e^{-z^{-3}} \dots$$

$$= \left[\sum_{n=0}^{\infty} \frac{(-z^{-1})^n}{n!} \right] \cdot \left[\sum_{n=0}^{\infty} \frac{(-z^{-2})^n}{n!} \right] \dots$$

$$= \left(1 - z^{-1} + \frac{z^{-2}}{2!} - \frac{z^{-3}}{3!} + \frac{z^{-4}}{4!} - \dots\right) \cdot \left(1 - z^{-2} + \frac{z^{-4}}{2!} - \frac{z^{-6}}{3!} + \dots\right) \dots$$

$$= 1 - z^{-1} + \left(\frac{1}{2} - 1\right)z^{-2} + \left(-\frac{1}{3!} + 1 - 1\right)z^{-3} + \dots = 1 - z^{-1} - \frac{1}{2}z^{-2} - \frac{1}{6}z^{-3} - \dots$$

14.证明函数 $f(z) = \sin(z + \frac{1}{z})$ 用 z 的幂表示的洛朗展开式的系数为

$$C_n = \frac{1}{2\pi} \int_0^{2\pi} \cos n\theta \sin(2 \cos \theta) d\theta, n = 0, \pm 1, \pm 2, \dots$$

证明:

$$\text{设 } f(z) \text{ 在 } z_0 = 0 \text{ 处洛朗展开式为 } \sin\left(z + \frac{1}{z}\right) = \sum_{n=-\infty}^{+\infty} c_n z^n,$$

$$\text{其中 } c_n = \frac{1}{2\pi i} \oint_C \frac{\sin(z + \frac{1}{z})}{z^{n+1}} dz,$$

C 为绕 $z_0=0$ 的简单正向闭曲线, 不妨设 C 为: $|z|=1$

令 $z = e^{i\theta}$, 则 $dz = ie^{i\theta} d\theta$

$$\begin{aligned} c_n &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{\sin(e^{i\theta} + e^{-i\theta})}{e^{i(n+1)\theta}} ie^{i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} \frac{\sin(2\cos\theta)}{e^{in\theta}} d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{\sin(2\cos\theta)}{\cos n\theta + i\sin n\theta} d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \sin(2\cos\theta) \cdot (\cos n\theta - i\sin n\theta) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \sin(2\cos\theta) \cdot \cos n\theta d\theta - \frac{i}{2\pi} \int_0^{2\pi} \sin(2\cos\theta) \cdot \sin n\theta d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \cos n\theta \sin(2\cos\theta) d\theta, \quad n = 0, \pm 1, \pm 2, \dots \end{aligned}$$

奇函数, 在
[0, 2\pi] 积分为0

